

2.5 Canonical and Standard Forms

2.5.1 Minterms and Maxterms

Table 2-3 Minterms and maxterms for three binary variables

<u>Minterms</u>				<u>Maxterms</u>	
x	y	z	Term Designation	Term	Designation
0	0	0	xyz' m_0	$x+y+z$ M_0	
0	0	1	$xy'z$ m_1	$x+y+z'$ M_1	
0	1	0	$x'yz$ m_2	$x+y'+z$ M_2	
0	1	1	$x'yz'$ m_3	$x+y'+z'$ M_3	
1	0	0	$x'y'z$ m_4	$x'+y+z$ M_4	
1	0	1	$x'y'z'$ m_5	$x'+y+z'$ M_5	
1	1	0	$x'yz'$ m_6	$x'+y'+z$ M_6	
1	1	1	xyz m_7	$x'+y'+z'$ M_7	

- A binary variable may appear either in its normal form (x) or in its complement form (x').
- Consider two binary variables x and y combined with an AND operation. There are 4 possible combinations: xy' , xy , xy' and xy .
- Each of these four AND terms is called a minterm or a standard product.
- n variables can also be combined to form 2^n minterms.
- The binary numbers from 0 to $2^n - 1$ are listed under the n variables. Each minterm is obtained from an AND term of the n variables, with each variable being primed if the corresponding bit of the binary number is a 0 and unprimed if a 1.
- In a similar fashion, n variables forming an OR term, with each variable being primed or unprimed are called maxterms or standard sums.
- Each maxterm is obtained from an OR term of the n variables, with each variable being unprimed if the corresponding bit is a 0 and primed if a 1.
- Each maxterm is the complement of its corresponding minterm and vice-versa.